Interoperability Test

2025-07-29-11-08-27

Objective

The Candela Interoperability test is designed to measure an Access Point's client performance when handling different amounts of Real clients like android, Linux, windows, and IOS. The test allows the user to increase the number of clients in user-defined steps for each test iteration and measure the per-client throughput for each trial. Along with throughput other measurements made are client connection times, Station 4-Way Handshake time, DHCP times, and more. The expected behavior is for the AP to be able to handle several stations (within the limitations of the AP specs) and make sure all Clients get a fair amount of airtime both upstream and downstream. An AP that scales well will not show a significant overall throughput decrease as more Real clients are added.

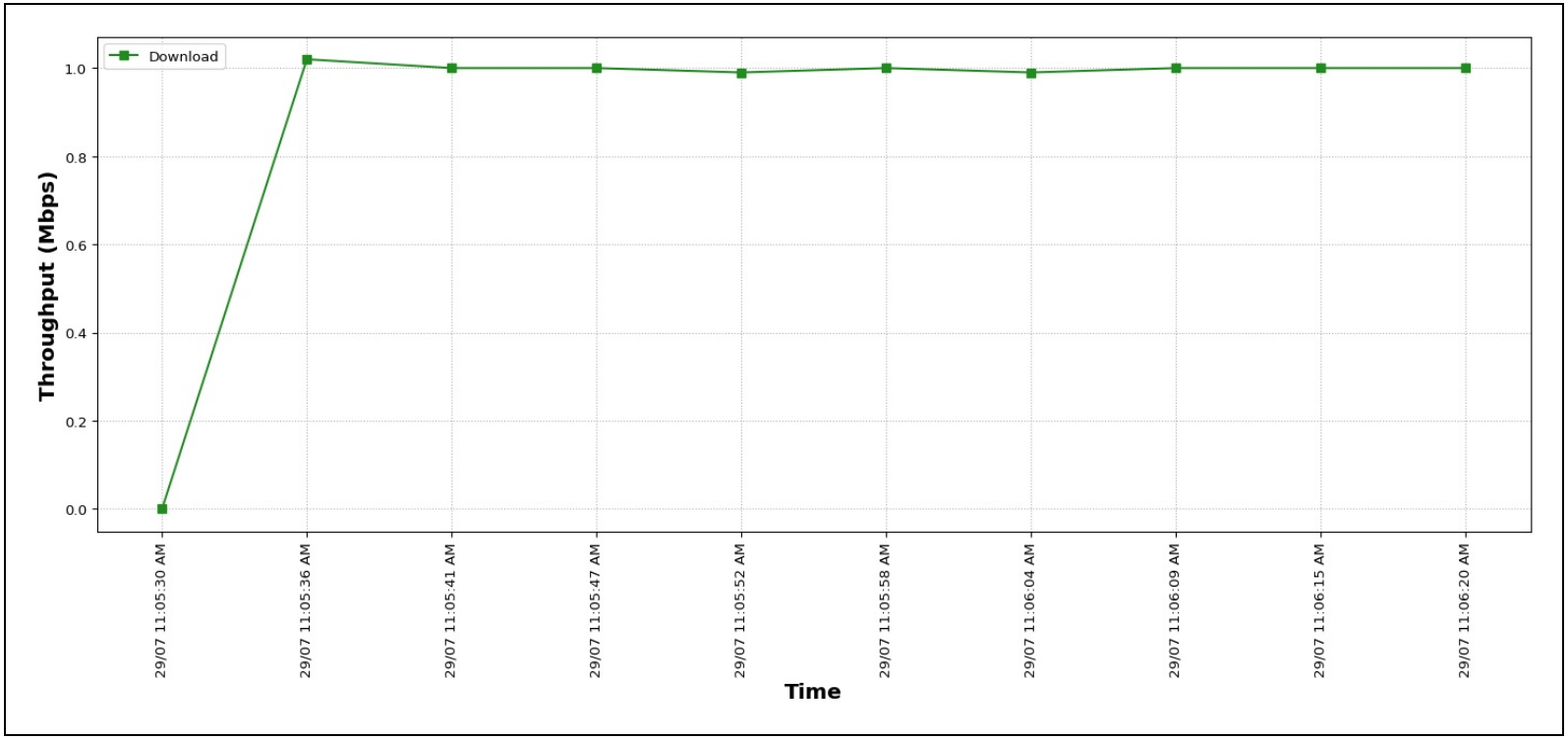
Input Parameters

The below tables provides the input parameters for the test

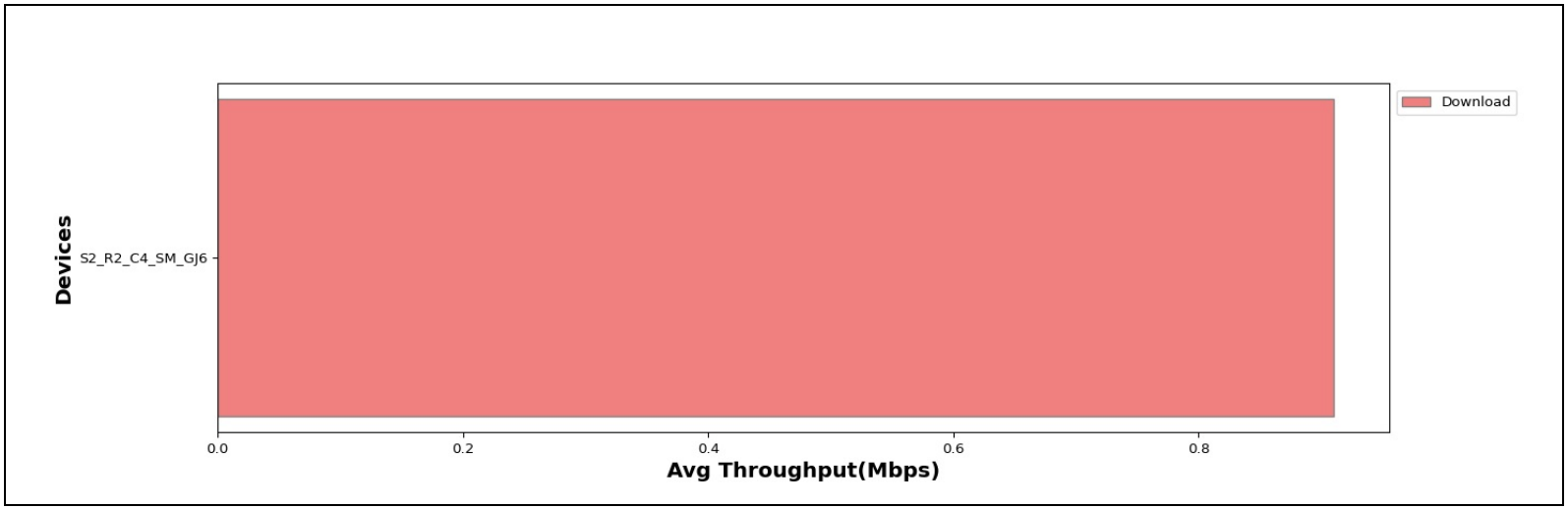
Test Configuration	Test name	None
	Device List	S2_R2_C4_SM_GJ6(Android), DesktopLatitude5490(Windows), mac-11.local(Mac)
	No of Devices	Total(3) Android(1) Windows(1) Mac(1)
	Traffic Duration in minutes	3.0
	Traffic Type	UDP
	Traffic Direction	Download
	Upload Rate(Mbps)	0.0Mbps
	Download Rate(Mbps)	1.0Mbps

1. Test On Device S2_R2_C4_SM_GJ6:

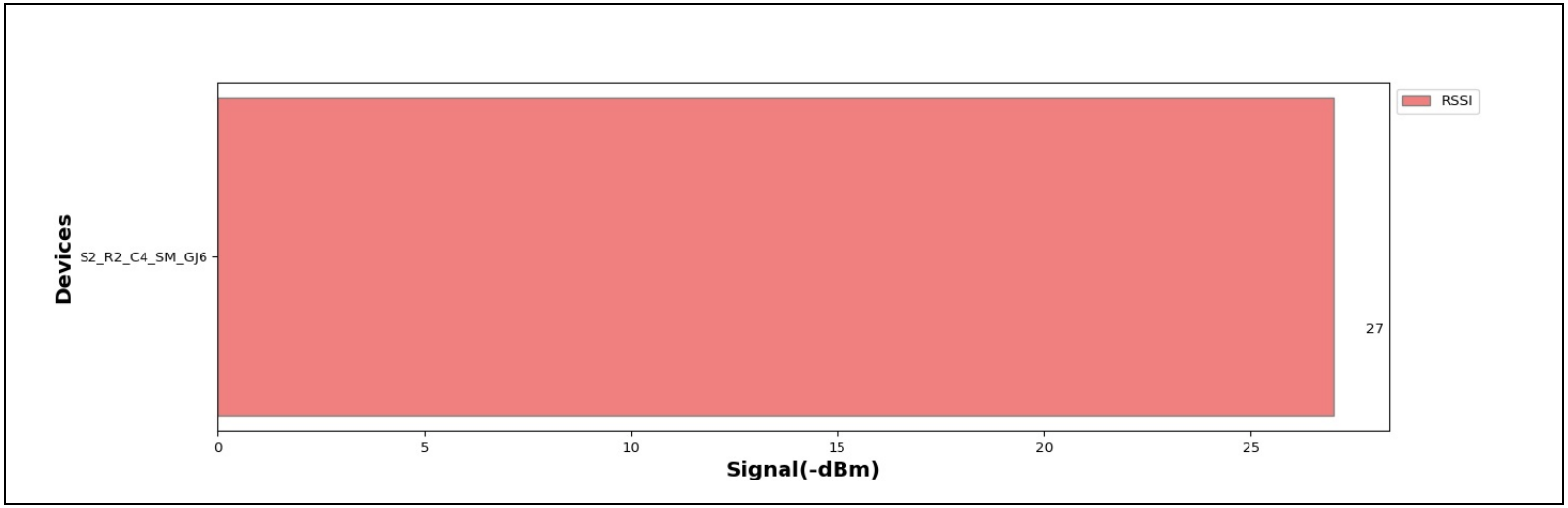
Real Time Throughput: Achieved Throughput: Download: 0.91 Mbps



Per Client Avg-Throughput



RSSI Of The Clients Connected



Detailed Result Table

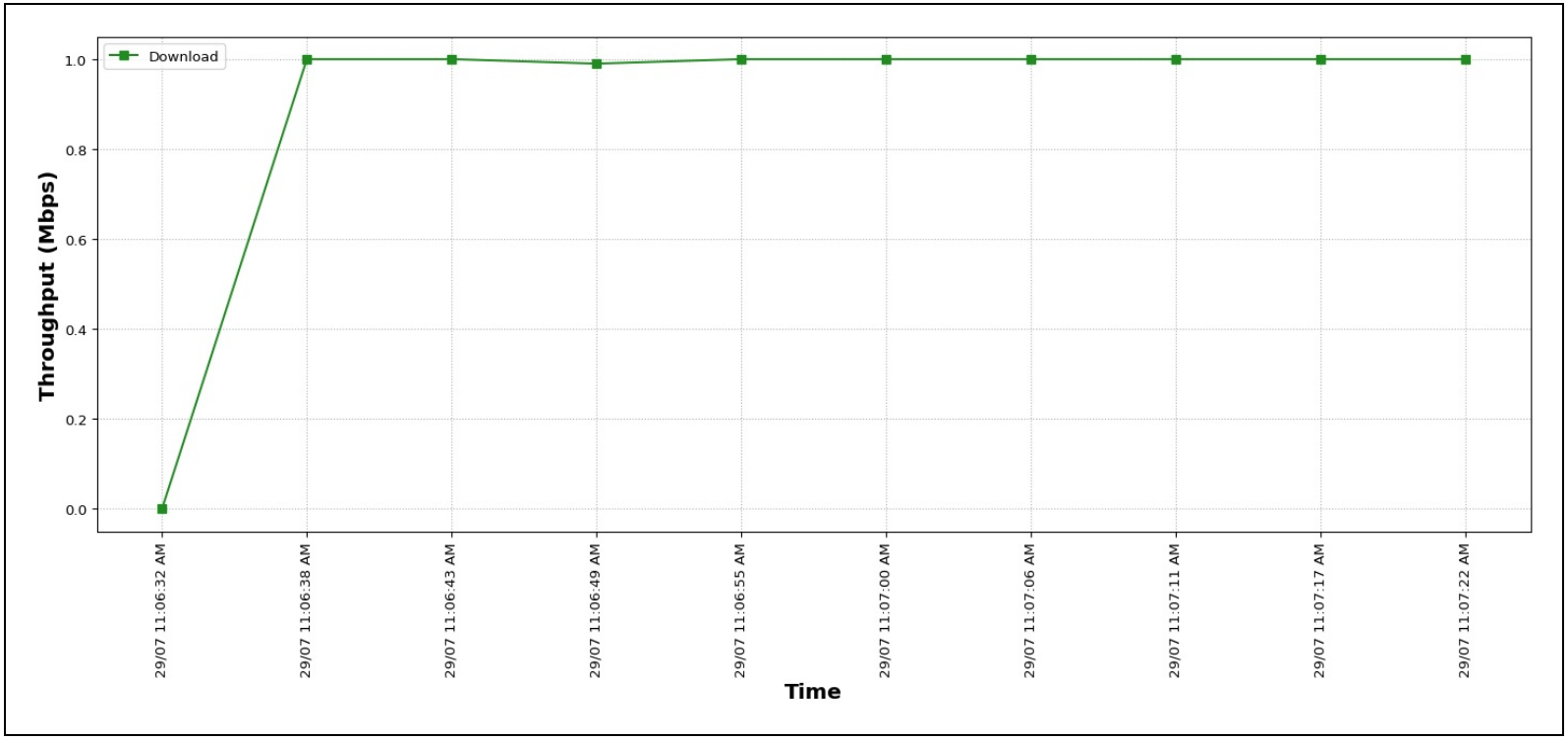
The below tables provides detailed information for the throughput test on each device.

Device Type	Username	SSID	MAC	Channel	Mode	Offered download rate	Observed Average download rate	Offered upload rate	Observed Average upload rate	RSSI	Average Rx Drop A%	Average Rx Drop B%
Android	S2_R2_C4_SM_GJ6	xiab_2G_WPA2	fe:80:53:eb:57:18	6	AUTO 20	1.0Mbps	0.91 Mbps	0.0Mbps	0 Mbps	-27 dbm	0.01	0.0

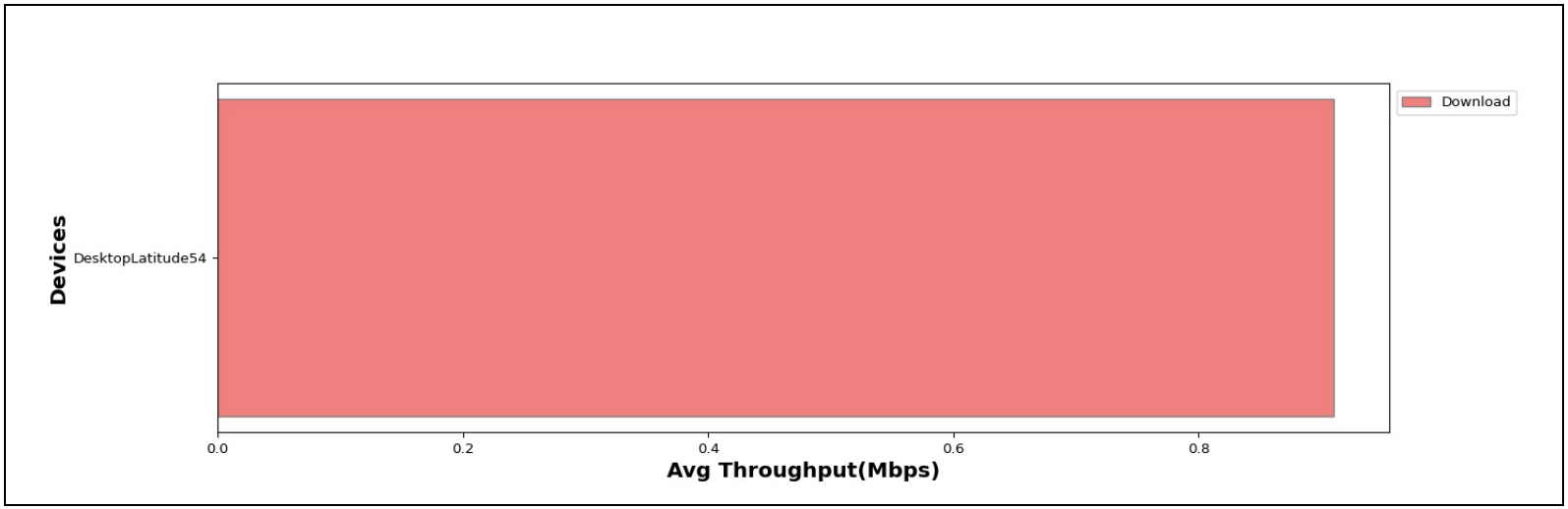
---

2. Test On Device DesktopLatitude5490:

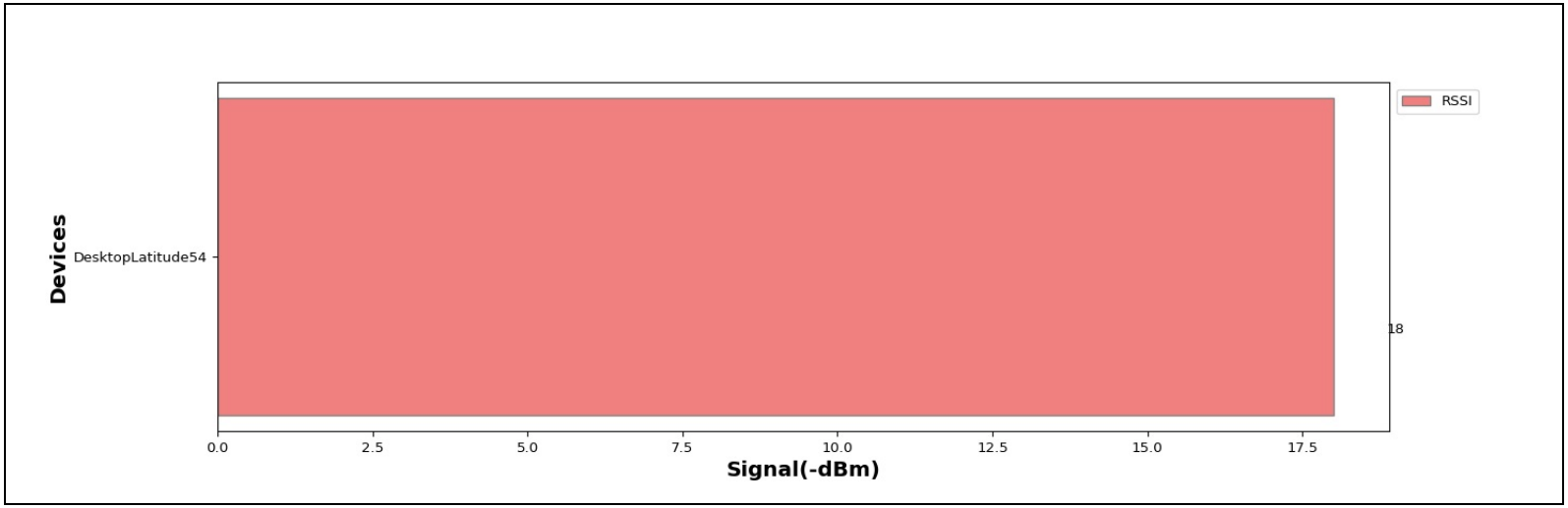
Real Time Throughput: Achieved Throughput: Download: 0.91 Mbps



Per Client Avg-Throughput



RSSI Of The Clients Connected



Detailed Result Table

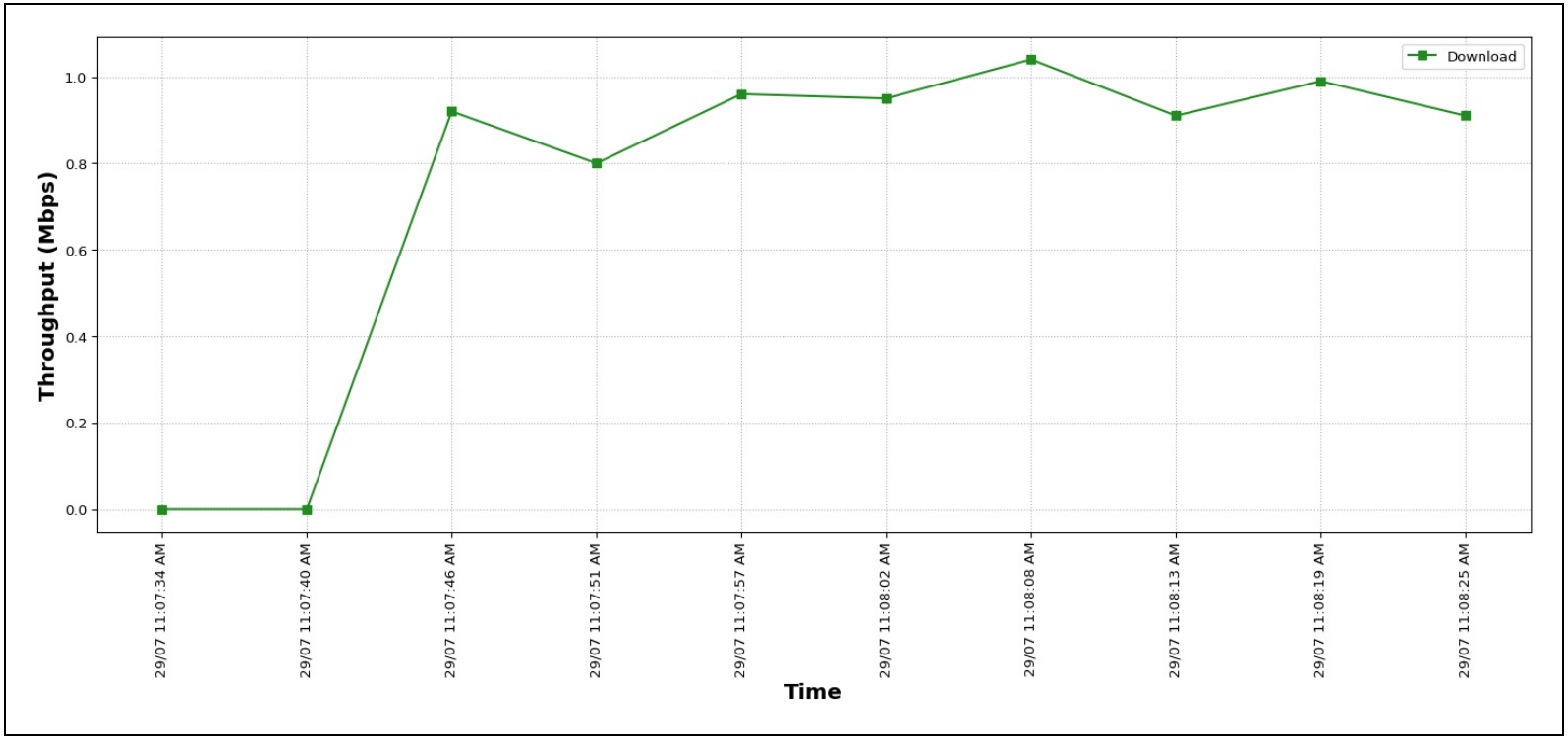
The below tables provides detailed information for the throughput test on each device.

Device Type	Username	SSID	MAC	Channel	Mode	Offered download rate	Observed Average download rate	Offered upload rate	Observed Average upload rate	RSSI	Average Rx Drop A%	Average Rx Drop B%
Windows	DesktopLatitude5490	xiab_2G_WPA2	dc:8b:28:28:d7:64	6	802.11abgn 20 1x1	1.0Mbps	0.91 Mbps	0.0Mbps	0 Mbps	-18 dbm	0.0	0.0

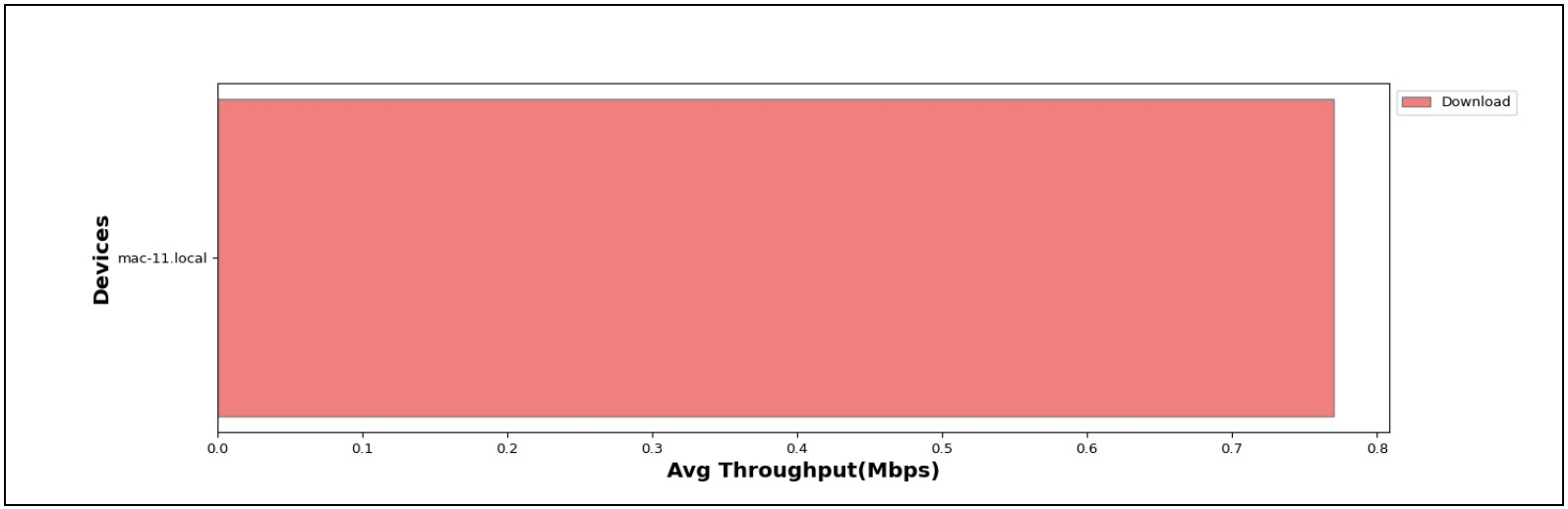
---

3. Test On Device mac-11.local:

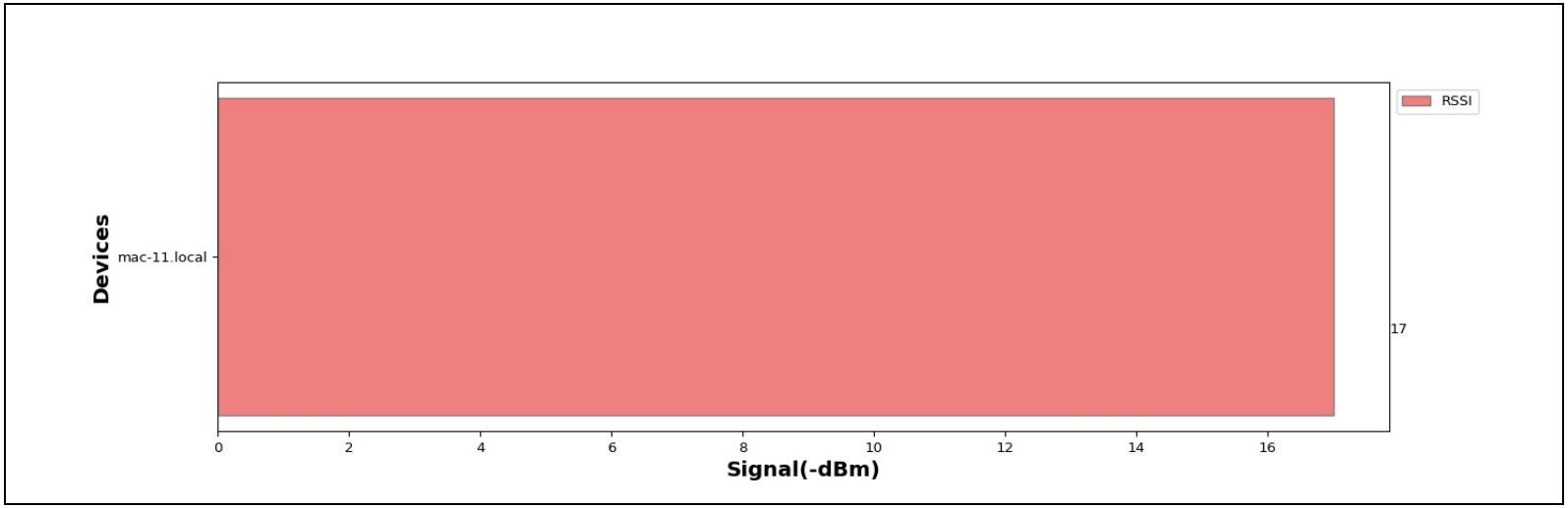
Real Time Throughput: Achieved Throughput: Download: 0.77 Mbps



Per Client Avg-Throughput



RSSI Of The Clients Connected



Detailed Result Table

The below tables provides detailed information for the throughput test on each device.

Device Type	Username	SSID	MAC	Channel	Mode	Offered download rate	Observed Average download rate	Offered upload rate	Observed Average upload rate	RSSI	Average Rx Drop A%	Average Rx Drop B%
Mac	mac-11.local	xiab_2G_WPA2	ac:bc:32:77:dc:99	6	802.11abgn-BE 20 1x1	1.0Mbps	0.77 Mbps	0.0Mbps	0 Mbps	-17 dbm	3.92	0.0